

Center of Excellence (CoE)

1. INTRODUCTION:

In alignment with the national programme, APIS as part of department of ITE&C, Govt of AP in order to promote the new & emerging technologies is in the process of setting up of CoEs' which encompasses various emerging technologies like AR & VR, Autonomous & Robotics, Green & Sustainable Energy, Energy Storage Technologies. The initiative would create a foundation and a seamless ecosystem from education, human resource, R&D, technology development, innovation to commercialization. It will generate Indian Start-ups, Intellectual Property (IP), Licensing & Monetization, jobs, enhance skills and world class product development resulting in technology driven economic growth.

2. BACKGROUND

APIS after an elaborate exercise with the various stakeholders from public & private sectors and the Startup Ecosystem on this initiative has come up with an implementable idea of CoEs'. APIS had come up with requirements in terms of what would be beneficial to the student community, startups and the incubators as well as the Industries who would like to consume the services being built by these Incubators in the emerging technologies.

3. PROPOSAL

After all the deliberations, a Hub (APIS), Spoke (Incubators) and spike (Colleges) structure is considered an effective model and the same has been finalised. Following are the key CoEs' envisioned as a part of the initiative.

1. AR & VR
2. Autonomous
3. Sustainable Energy
4. Energy Storage Technologies
5. 5G Use Cases & Stack Development

4. Management Structure

The Master Hub at APIS, Visakhapatnam along with 26 incubators (spoke) spread across Andhra Pradesh are supposed to work closely with the spike institutions (Colleges) as mentioned in the detailed proposals of the respective Hub to address some of the critical problems of national importance in the emerging technologies space. National level use-cases could be identified and provided to these incubators for implementation and pilots. This would help in giving the directed applied research problems to the CoE rather than researchers themselves take some research problems as per their choice.

In the proposed model the incubators are the core for implementation of the emerging technology CoEs. The CoEs are being executed by the APIS through a Section 8 Company constituted for the purpose (or existing Section 8 Company would cater to these as well by the APIS). The Board includes Partner Implementing Agency, Knowledge Partner, Industry Partner, Funding Agency, and Technology Partner. It is most important that either at the APIS level or at the Management Board level to see that the CoEs do not work in silos and work in coherence achieve national initiatives along with central/state government/sectoral organisations.

CoEs will have full autonomy in devising their internal processes and procedures for achieving the targets/deliverables. Each CoE will have full financial and administrative powers, including approvals to re-appropriation of the budget within the ceiling of sanctioned budget; hiring of the appropriate manpower as per industry standards; sign Memorandum of Understanding (MoU) with national and International institutions and industry; approve Collaboration foreign ; partner with industry; receive and give projects in the domain areas of Hub-Spoke-Spike to academia, R&D institutions, Industry and other funding agencies.

CoEs Established (Supported by APIS)

S.No	CoE	Place	Rs (In Crore)
1.	CoE IoT & AI (NASSCOM)	Visakhapatnam	5 (22.6 Cr for 5 Yrs)
2.	A HUB (Andhra University)	Visakhapatnam	10 (15 Cr for 3 Yrs)
3.	KALPATARU STPI (Industry 4.0)	Visakhapatnam	20.32
4.	AMTZ (Medtech)	Visakhapatnam	-
5.	CoE Maritime & Ship Building	Visakhapatnam	26
6.	STPI – Fablab (Proposed)	Vijayawada	6
7.	SKU AIC (Rural Incubation Centre)	Ananthapur	5.2 (10.7 Cr for 5 Yrs)

CoEs Proposed

S.No	CoE	Partner	Rs (In Crore)
1	AR & VR (Meta)	Microsoft	Hub - 5 Spoke - 5
2	Autonomous	Nasscom	
3	Energy Storage	NSTL	
4	Sustainable Energy	IIFE, Visakhapatnam	
5	5G Use Cases	GVP	

1.CoE - AR & VR (Meta)

APIS has come up with a unique initiative where they want to set up Centre of Excellences across different technologies with Meta being one of the key technologies.

There has been elaborate exercise that has been done with various stakeholders from public & private sectors in the Startup Ecosystem on this initiative.

As a part of the Brain storming Sessions APIS had come up with requirements in terms of what would be beneficial to the student community and the incubators as well as the Industries who would like to consume the services being built by these Incubators.

As part of the key items that have been identified are:

1. Technical Partner
2. Funding
3. Infrastructure
4. On boarding the Incubators

In this context, APIS have initially discussed with Microsoft which involves Key leadership, Microsoft has attended the meeting and they have approved in principle agreed to use the logo and provide curriculum and much customized trainings to the students.

Microsoft also appreciated the way the entire proposal is put together and they thought this is the right approach to build talent pool in the segment after that APIS had a series of conversations with multiple companies to understand the lab requirements and some study has been done to see infrastructure requirements.

An elaborate meeting has been called for during the visit of Director General, STPI & Industrial body ITAAP all been involved and they have been presented with this and everyone responded positively telling that this is the major initiative that can add a lot of value to the Ecosystem and the participants of the CoE.

After that STPI also appreciated and where in APIS had pitched the STPI to fund the initiative and in parallel APIS also showcase the Incubators by calling for a meeting with all the Incubators in the state. The incubators have expressed their willingness to be a part of this initiative and the following are the financials associated with the setup.

CoE – Metaverse Model

The CoE – Metaverse flow chart model is shown below

CoE – HUB
Regional HUB (Incubator)
SPOKE (Colleges)

Cost Projection (Tentative)

Find below the phases of Cost Projections

	Item	Rs (In Cr)
Phase 1	CoE - Hub	5 Cr
	Regional Hub (Incubators) 20 Satellite centers @ 25 lakh per center	5 Cr
Phase 2	Spoke (Colleges) 200 Engg colleges @ 10 lakh per college	20 Cr
	500 PG/Degree Colleges @5 lakh per college	25 Cr
	Total Cost	55 Cr

Revenue Generation – Shared

Described all the Activities & Revenue Generation in below table

Certifications (6/12/24 month Programs) 1. 200/234 Engg Colleges @1000 students/ college @ INR 2000 per certificate 2. 500 Other Colleges @100 students/college @ INR 500 per certificate	1. INR 40 Crore 2. INR 25 Crore
Skill Dev (Short & Long term) 1. 25000 @10000 per candidate	INR 25 Crore
Govt projects 1. Education/Smart cities/Healthcare/Industries/Defence	INR 10 Crore
Science Fairs 1. 250 Schools @ 10000 per school	INR 25 Lakhs
Internships/ Live Projects 1. 5000 /50000 students @5000 per student	INR 25 Lakhs
Industry Consulting services	INR 10 Crore
GRAND TOTAL (Per Year)	INR 110.5 Crore EBIDTA 44 Crore (40%)

Revenue Model

The proposed revenue model is given below

Hub (APIS)	50 %
Regional Hub (Incubators)	30 %
Spoke (Colleges)	20 %

There has been a constant exercise to evaluate various vendors who can supply the infrastructure, Technology and execution capability.

2.CoE - Sustainable Energy

APIS has come up with a unique initiative where they want to set up Centre of Excellences in different sectors where sustainable energy being one of the key sectors.

There has been elaborate exercise that has been done with various stakeholders from public & private sectors in the Startup Ecosystem on this initiative.

As a part of the Brain storming Sessions APIS had come up with requirements in terms of what would be beneficial to the state government, student community and the incubators as well as the Industries who would like to consume the services being built by these Incubators.

As part of the key items that have been identified are:

1. Industry/Technical Partner
2. Funding
3. Infrastructure
4. On boarding the Incubators

In this context, APIS has proposed to discuss with the following industries, knowledge partners and institutions to explore partnerships.

- a) HPCL - Hindustan Petroleum Corporation Limited
- b) IIPE - Indian Institute of Petroleum and Energy
- c) APMB - Andhra Pradesh Maritime Board
- d) GAIL - Gas Authority of India Limited
- e) IIT Delhi - Indian Institute of Technology Delhi

An elaborate meeting has been called for during the visit of Dy Chief Executive Officer, Andhra Pradesh Maritime Board (APMB), presentation received a positive reply appreciated that this is the major initiative that can add a lot of value to the Ecosystem and the participants of the CoE.

CoE – Sustainable Energy Flow Chart

The CoE – Sustainable Energy flow chart model is shown below

CoE – HUB
Regional HUB (Incubator)
SPOKE (Colleges)

Financial Outlay (Tentative)

S.NO	Activities	Sub-Activities	Budget in INR Cr
1.	Infrastructure (physical & virtual)	State of art ecosystem including (R&D lab)	4.00
		Man-hour & Manpower	
2.	Administrative	Office Operational and etc	1.00
3.	Promotional Activities	Industry innovation Activities and Exhibitions	0.50
4.	Miscellaneous Expenses	0.50	
Total			6.00

Revenue Outlay (Tentative)

S.NO	Activities	Sub-Activities	Revenue in Cr
1.	Industrial Consultancy	Govt. Projects (Infra integrity analysis)	10.00
		Private & Corporate Projects (DPRs)	
2.	Evaluations	Energy Audit	5.00
3.	R&D Activities	Research & Innovation	2.50
4.	Project evaluations	Project proposal	1.00
5.	Skill Development	FDP/MDP/training Students	2.50
6.	Certifications & Validation		2.00
7.	Rankings & Ratings		2.00
8.	Carbon trade off	Carbon Aggregation of green energy projects	2.00
9.	ToT & Commercialization		1.00
10.	Hackathons/Cohorts/Master Class/Buyer-Seller		0.50
Total			28.50

3. CoE - Autonomous:

Recent geopolitical developments, such as the refusal of Bayraktar combat drones to India, have brought to the front the need for self-reliance on critical technologies such as Autonomous Tech. One such technology is Autonomous Technologies that ranks in the top 10 emerging technologies for advanced economies such as USA, U.K., Japan and Australia as they have dual use in both civil and military applications. Autonomous Tech includes several foundational Tech that include GPS, LIDAR, SONAR, RADAR AND Spectral Cameras on the Hardware end, Object avoidance and SLAM algorithms on the software end and electric transmission technologies.

India contributes to over 35% of the Autonomous software development and has moved up the value chain on the Electric Vehicle technologies other than advances on the Underwater Autonomous missiles that DRDO has developed. An Autonomous CoE can build capabilities/IP across Hardware, Software and EV end and can integrate the siloed efforts across the Civil and Military domains to enable India solve for its Manufacturing (Autonomous Mobile Robots), Maritime (Underwater guided vehicles) and Aerospace & Defence (Underwater Missiles and long range combat UAVs).

4. CoE - Energy Storage Technologies:

Energy Storage System (ESS) is fast emerging as an essential part of the evolving clean energy systems of the 21st century. Energy storage represents a huge economic opportunity for India.

India's strengths in IT and manufacturing, its entrepreneurial and dynamic private sector, and its visionary public and private sector leadership will be key factors in realizing energy storage goals.

Energy storage technologies can be created into five major categories such as batteries, thermal, mechanical storage, hydrogen and pumped hydropower. Storage technologies help in building resilient infrastructure and bring cost savings to utilities and consumers.

- Batteries – a range of electrochemical storage solutions, including advanced chemistry batteries, flow batteries, and capacitors

- Thermal – capturing heat and cold to create energy on demand or offset energy needs
- Mechanical Storage – other innovative technologies to harness kinetic or gravitational energy to store electricity
- Hydrogen – excess electricity generation can be converted into hydrogen via electrolysis and stored
- Pumped Hydropower – creating large-scale reservoirs of energy with water

The energy storage landscape is well split among organizations with various technologies and these technology companies are broadly based out of Japan, China, Korea, US and Germany.

India has committed to increase its share of non-fossil fuel-based generation sources to 45% by 2030 which necessitates a demand for flexibility in power systems. The ‘Power for All’ target of 24x7 electricity for all by 2019 created an increase in power requirement and a need to balance the supply and demand of electricity. Energy storage will play a crucial role in increasing the system’s overall flexibility by serving multiple grid applications. The recent developments in the Electric Vehicle (EV) sector and its ambitious targets will only increase the demand for energy storage systems.